SHRIMP DISEASE FIELD GUIDE

Comprehensive Reference for Aquaculture Farmers & Field Technicians

15 Major Diseases | Symptoms, Causes, Illustrations, Impact & Prevention

Focus Region: Andhra Pradesh (Nellore, Godavari & Krishna Districts)

Introduction

Shrimp aquaculture in Andhra Pradesh is a critical industry, supporting hundreds of thousands of livelihoods along the coastal belts of Nellore, East and West Godavari, and Krishna districts. However, disease outbreaks remain the single largest threat to productivity and profitability. This field guide documents the 15 most clinically significant diseases affecting Penaeus vannamei (Pacific white shrimp) and Penaeus monodon (black tiger shrimp).

Each disease profile includes the causative agent, key symptoms, behavioral changes, economic impact, prevention and management strategies, and a scientific illustration showing the key visual signs of each disease to aid rapid field identification.

Note: Illustrations are scientific diagrams created for educational purposes. Definitive diagnosis must be confirmed by PCR testing or histopathology.

Quick Reference Summary Table

#	Disease Name	Type	Key Sign	Risk
1	White Spot Syndrome Virus (WSSV)	Viral	White spots on shell, redness	**Critical**
2	EHP (E. hepatopenaei)	Parasitic	Extreme growth retardation	**Critical**
3	Running Mortality Syndrome (RMS)	Multi-factorial	Slow ~1% daily deaths	**High**
4	White Feces Syndrome (WFS)	Mixed	White floating feces strings	**High**
5	Black Gill Disease	Environmental/Bacterial	Dark/black gills	**Moderate**
6	IMNV (Myo)	Viral	Opaque white tail muscles	**High**
7	Covert Mortality Nodavirus (CMNV)	Viral	Empty stomachs, sink on death	**High**

#	Disease Name	Type	Key Sign	Risk
8	Decapod Iridescent Virus 1 (DIV1)	Viral	Iridescent glow, white head	**Critical**
9	Vibriosis	Bacterial	Bioluminescence, redness	**High**
10	IHHNV / Runt-Deformity Syndrome	Viral	Bent rostrum, stunted growth	**Moderate**
11	Loose Shell Syndrome (LSS)	Environmental	Shell detached from meat	**Moderate**
12	Hepatopancreatic Parvovirus (HPV)	Viral	Pale, shrunken hepatopancreas	**Moderate**
13	Yellow Head Disease (YHD)	Viral	Yellow cephalothorax, sudden die-off	**High**
14	Luminescent Vibriosis	Bacterial	Greenish-blue glow at night	**High**
15	Tail Necrosis (Rotten Tail)	Bacterial	Black, eroding tail fan	**Moderate**

Detailed Disease Profiles with Illustrations

1. White Spot Syndrome Virus (WSSV) [Critical Risk]

Cause / Type	Viral (Whispovirus). Highly contagious; spreads through infected water, raw crustacean feed, equipment, and carrier organisms including crabs and wild shrimp.
Symptoms	White calcium deposits (2-5 mm spots) on inner carapace and tail; reddish-pink body discoloration; sudden catastrophic mortality (can wipe entire pond in 3-10 days).
Behavior	Infected shrimp congregate near pond surface and edges; lethargic; cessation of feeding.
Economic Impact	Can cause 100% mortality in untreated ponds within 7-10 days. Complete crop loss and mandatory pond disinfection required.
Prevention / Management	Use SPF-certified postlarvae. PCR screen all broodstock. Eliminate carrier crabs and wild shrimp. Apply lime (CaO) at 2000 kg/ha for disinfection post-outbreak.

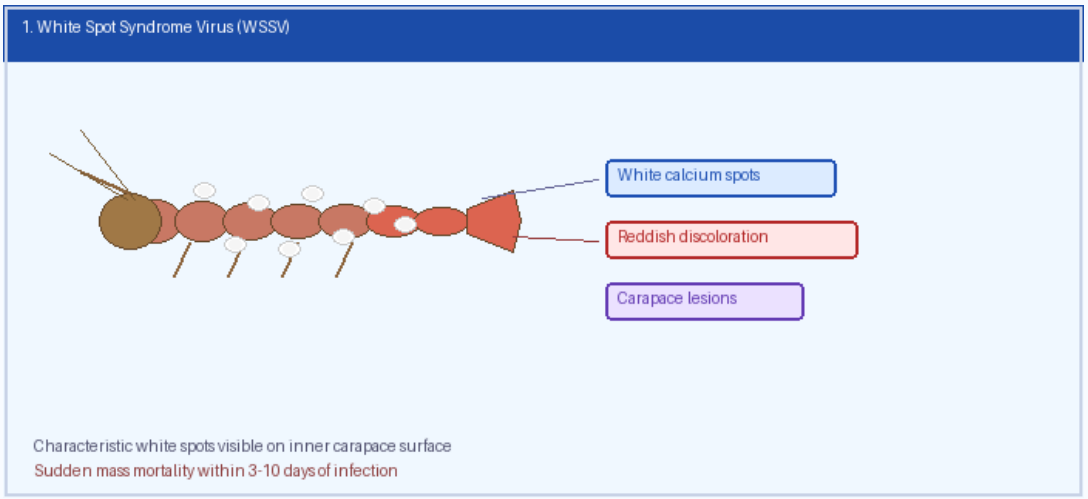

Fig 1: WSSV-infected shrimp showing white calcium spots on carapace and reddish body discoloration

2. Enterocytozoon hepatopenaei (EHP) [Critical Risk]

Cause / Type	Microsporidian intracellular parasite infecting hepatopancreas tubule epithelial cells. Transmitted through contaminated sediment, water, and infected feed.
Symptoms	No acute mass mortality. Extreme growth retardation and severe size variation ('size variation syndrome'). Some shrimp reach 18g while cohabitants stay at 3-5g by Day 90. Pale, atrophied hepatopancreas.
Behavior	Lethargic; fall behind in feed consumption; more susceptible to secondary infections.
Economic Impact	Production costs increase 20-30%. Surveys in Nellore and Godavari districts (2025-2026) report over 60% infection prevalence in some farms.
Prevention / Management	PCR-based detection before stocking. Dry pond completely between cycles; apply 300 kg/acre of lime to kill spores. Avoid reusing infected pond water.
Current Status	Infection rates over 60% reported in Nellore and Godavari districts (2026), forcing early harvests.

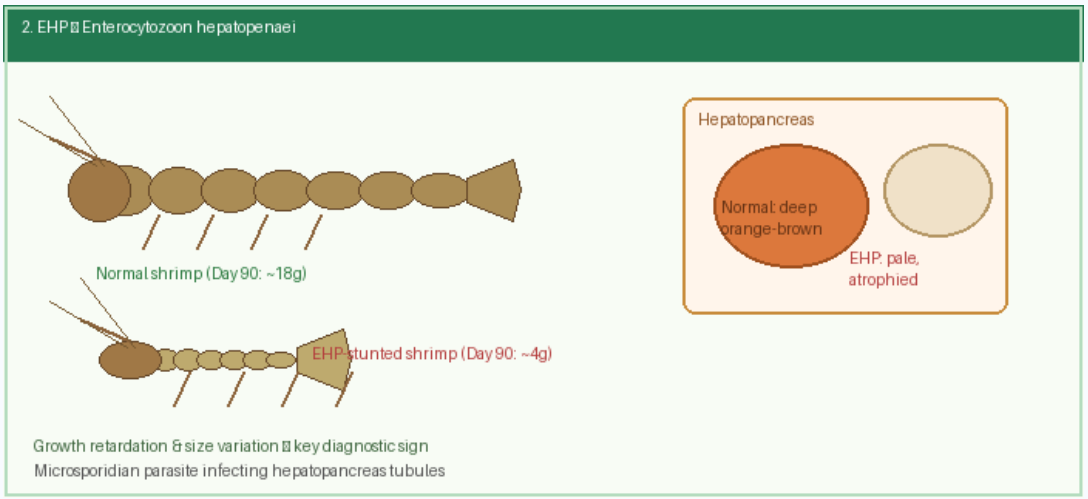

Fig 2: EHP size variation — normal shrimp (top) vs severely stunted EHP-affected shrimp (bottom) at Day 90

3. Running Mortality Syndrome (RMS) [High Risk]

Cause / Type	Multi-factorial: EHP co-infection, Vibrio spp. (especially V. parahaemolyticus), environmental stress, and suboptimal nutrition.
Symptoms	Continuous low-level daily mortality (~0.5-2%). White/yellow floating feces. Shrunken, pale hepatopancreas. Slow-growing undersized shrimp.
Behavior	Slow drain of losses — psychologically difficult for farmers to act decisively until cumulative damage is severe.
Economic Impact	Cumulative mortality of 30-60% over a 90-day cycle common in affected ponds.
Prevention / Management	Regular PCR monitoring for EHP. Strict water quality management, reduce organic load. Use premium feed with immunostimulants. Act immediately if check trays show >0.5% daily loss.

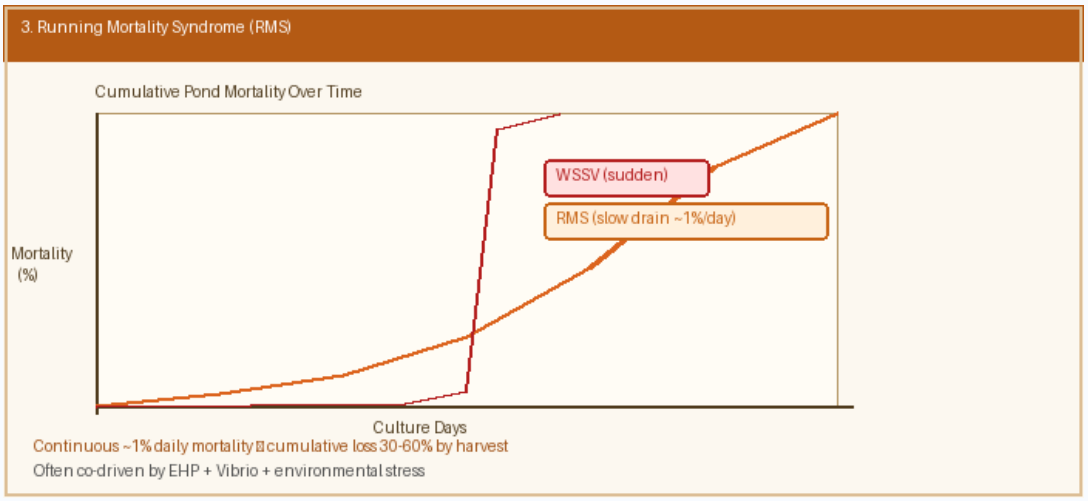

Fig 3: Mortality curve comparison — WSSV (sudden spike) vs RMS (slow continuous drain ~1% per day)

4. White Feces Syndrome (WFS) [High Risk]

Cause / Type	Mixed etiology: EHP infection combined with opportunistic Vibrio and Gregarine parasites. High organic loads and poor water quality accelerate onset.
Symptoms	White, worm-like tubular fecal strings floating on pond water surface — characteristic diagnostic sign. Pale hepatopancreas on dissection. Reduced feeding rate and slow growth.
Behavior	Reduced appetite; sluggish movement; shrimp may congregate near feed trays even when feed is removed.
Economic Impact	Growth rates drop 20-30%, extending culture period and increasing FCR. Strongly associated with EHP.
Prevention / Management	Reduce organic load (stop feeding 1-2 days, apply probiotics). PCR test for EHP. Apply vitamin-mineral supplements. Increase aeration and water exchange.

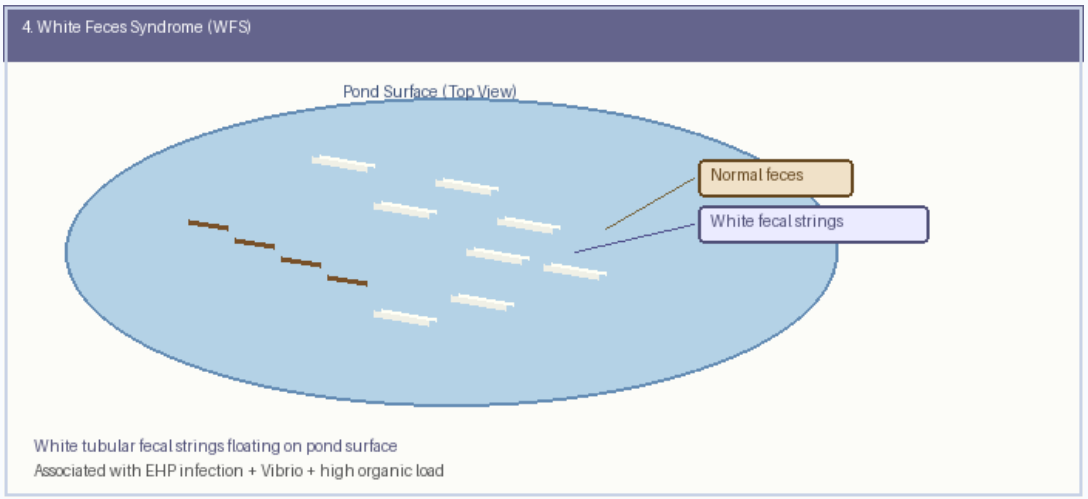

Fig 4: Top-down view of pond surface showing diagnostic white tubular fecal strings floating on water (vs normal brown feces)

5. Black Gill Disease [Moderate Risk]

Cause / Type	Environmental and opportunistic. Caused by organic debris, silt, fungal growth (Fusarium spp.), and bacterial colonization on gill surfaces. Indicator of poor pond bottom quality.
Symptoms	Gills appear brown, dark brown, or jet-black. Necrotic gill tissue. Reduced respiratory efficiency. Affected shrimp may surface in severe cases.
Behavior	Shrimp move toward aerators seeking better-oxygenated water. Reduced feeding response.
Economic Impact	Reduces growth and feed conversion. Indicates systemic pond health deterioration.
Prevention / Management	Primary: increase aeration, apply zeolite (200 kg/acre) to adsorb ammonia and H2S. Use bottom-targeted probiotics. Reduce organic input. Address the root cause (excess sludge).

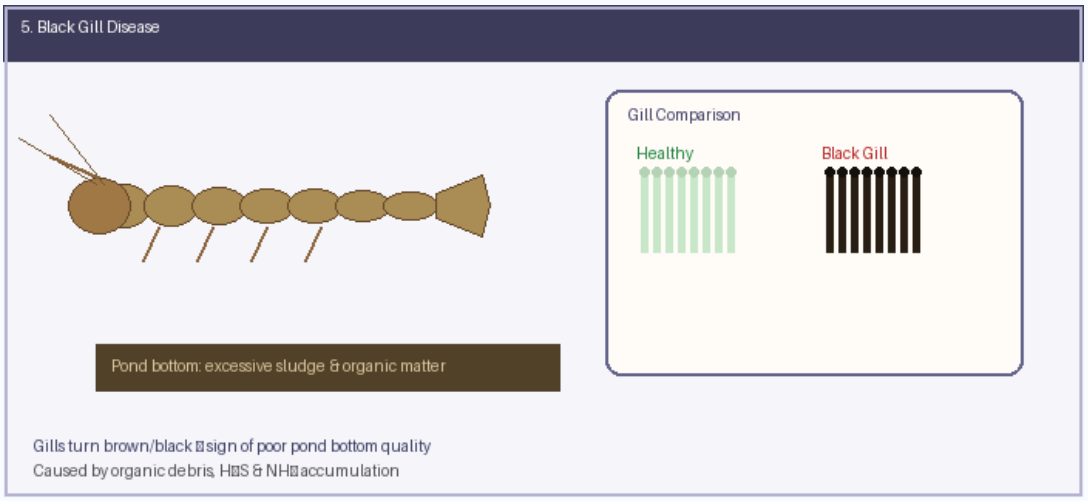

Fig 5: Gill comparison — healthy light-pink gills (left) vs melanized black gills (right) caused by organic debris accumulation

6. IMNV — Infectious Myonecrosis Virus ("Myo") [High Risk]

Cause / Type	Viral (Totiviridae). Infects skeletal muscle tissue. Transmitted through contaminated water, cannibalism of infected shrimp. Called 'Myo' by local farmers.
Symptoms	Opaque, white, cooked appearance of tail muscle segments visible through shell. In advanced stages, tail turns reddish ('cooked shrimp' appearance while alive). Mortality 40-70%.
Behavior	Hyperactive, jumpy behavior initially — erratic swimming bursts. Triggered by sudden temperature drops, salinity shifts, or handling.
Economic Impact	40-70% mortality in an outbreak cycle. Survivors often stunted and unsaleable. Increasing in Krishna and Godavari districts.
Prevention / Management	Avoid abrupt environmental changes. Maintain stable water temperature and salinity. Reduce stocking density during high-risk periods. PCR screen broodstock. No approved antiviral treatment.

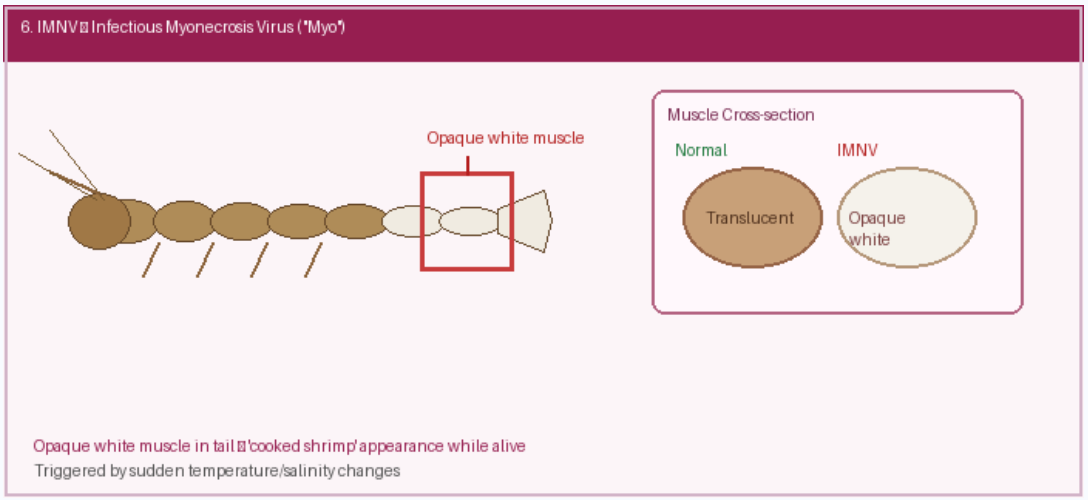

Fig 6: IMNV opaque white tail muscle (highlighted region). Cross-section shows normal translucent tissue vs necrotic white muscle

7. Covert Mortality Nodavirus (CMNV) [High Risk]

Cause / Type	Viral (Nodavirus). Emerging and underdiagnosed pathogen. 'Covert' refers to infected shrimp sinking on death, making mortality invisible at the surface.
Symptoms	Soft shell, slow growth, empty stomachs on inspection. Whitening of muscle tissue. Shrimp die at pond bottom — only detected during check-tray sampling.
Behavior	No distinctive surface behavioral sign. Internal check-tray sampling reveals dead shrimp with empty guts and soft exoskeletons.
Economic Impact	Highly underreported due to covert mortality. True losses may be significantly higher than surface observations indicate. Under surveillance in East Godavari district (2026).
Prevention / Management	Increase check-tray sampling to 3x daily. PCR screening for CMNV before and during culture. Maintain nutrition with immunostimulants (beta-glucan, Vitamin C).
Current Status	Increasing surveillance in East Godavari identifies CMNV as hidden cause of low productivity in 2026.

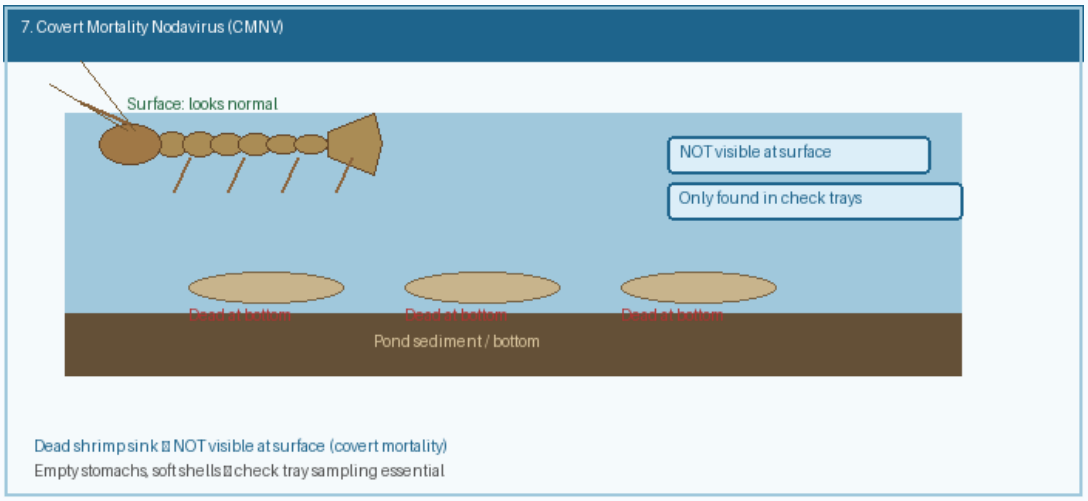

Fig 7: Pond cross-section showing covert mortality — surface appears normal, but dead/moribund shrimp accumulate at the pond bottom (invisible without check trays)

8. Decapod Iridescent Virus 1 (DIV1) [Critical Risk]

Cause / Type	Viral (Iridoviridae). Highly lethal emerging pathogen. Mandatory reportable disease in India. Affects both P. vannamei and P. monodon.
Symptoms	Iridescent blue or purple glow on body surface. Whitish area under carapace ('white head') due to hepatopancreas necrosis. Severe systemic infection.
Behavior	Rapid decline — from apparent health to severe mortality within days. Shrimp stop feeding abruptly.
Economic Impact	Up to 80% mortality within 14 days. Mandatory reportable disease in India — confirmed outbreaks must be notified to MPEDA/NACA.
Prevention / Management	Strict biosecurity; PCR test all imported seed for DIV1. Report confirmed outbreaks to authorities immediately. Disinfect with chlorine (30 ppm) before restocking. No approved treatment.

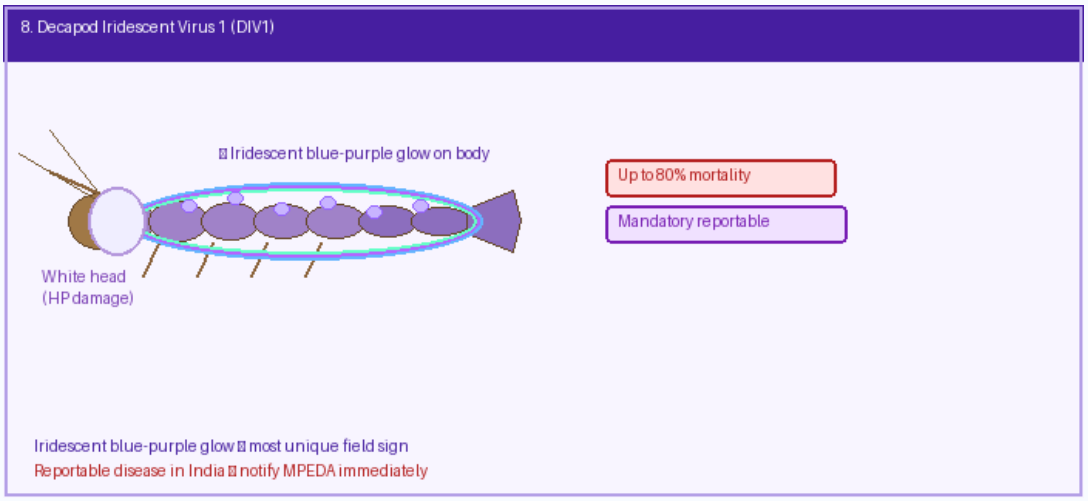

Fig 8: DIV1 infected shrimp showing iridescent blue-purple glow on body surface and whitened head area (hepatopancreas damage)

9. Vibriosis (Bacterial Septicemia) [High Risk]

Cause / Type	Bacterial. Caused by pathogenic Vibrio spp. (V. harveyi, V. parahaemolyticus, V. alginolyticus). Disease occurs when bacterial load overwhelms shrimp immunity under stress.
Symptoms	Reddish discoloration of body and appendages. Bioluminescence — pond water glows greenish-blue at night. Tail rot and erosion of uropods. Lethargy and reduced feeding.
Behavior	Bioluminescence most visible on moonless nights — one of the most recognizable field signs in shrimp farming.
Economic Impact	Can cause 30-80% mortality if unmanaged. Often a secondary infection complicating EHP, WSSV, or RMS. Common cause of hatchery mortality.
Prevention / Management	Bottom sludge management is the primary intervention. Apply probiotics (Bacillus spp.) to outcompete Vibrio. Reduce feeding during outbreak. Antibiotic use under veterinary supervision only.

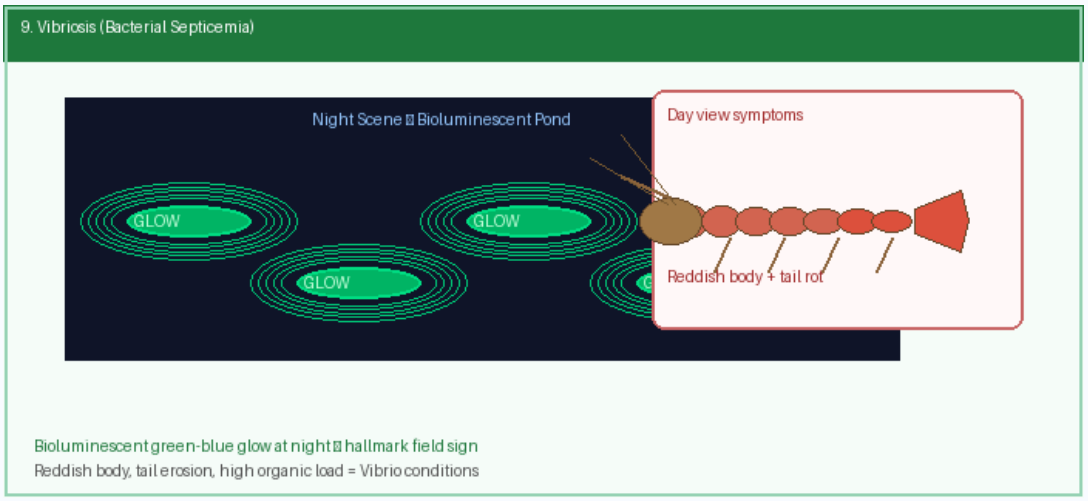

Fig 9: Bioluminescent Vibrio harveyi in pond water at night (left — greenish-blue glow) and day symptoms showing reddish body discoloration (right)

10. IHHNV — Runt Deformity Syndrome (RDS) [Moderate Risk]

Cause / Type	Viral (Parvovirus, Brevidensovirus genus). Infects hypodermal and hematopoietic tissue. Significant concern in improperly screened hatcheries.
Symptoms	Severely stunted growth ('runts'). Bent or twisted rostrum is the hallmark physical sign. Cuticular lesions and rough irregular shell texture. Rarely causes mass mortality.
Behavior	Affected shrimp appear behaviorally normal but are physically smaller and deformed.
Economic Impact	Since shrimp don't die, farmers may not notice until harvest — discovering 20-40% of crop is unsaleable due to size and deformity. Enormous feed and time investment wasted.
Prevention / Management	PCR screening of postlarvae before stocking is essential. Use IHHNV-free certified seed. No treatment once infected. Maintain strict hatchery biosecurity.

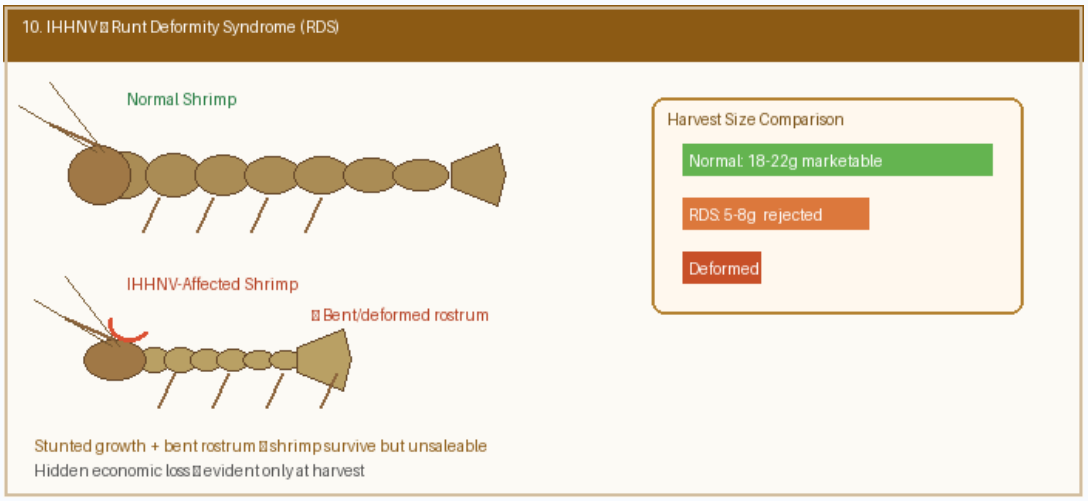

Fig 10: IHHNV comparison — normal full-size shrimp (top) vs stunted IHHNV-affected shrimp with characteristic bent rostrum (bottom). Harvest size impact chart shown right

11. Loose Shell Syndrome (LSS) [Moderate Risk]

Cause / Type	Environmental and nutritional. Linked to poor water quality, low dissolved oxygen, mineral imbalances (Ca/Mg deficiency), high organic loads. Most prevalent in low-salinity (<5 ppt) ponds.
Symptoms	Shell feels soft and detached from underlying muscle tissue — visible gap on pressing. Muscle tissue flabby, watery, pale. Mild lethargy.
Behavior	Highly vulnerable to cannibalism and secondary bacterial infection.
Economic Impact	Market price for affected batches drops 20-35%. Poor survival during transport and processing.
Prevention / Management	Supplement pond water with calcium ($CaCO_3$, dolomite) and trace minerals. Maintain salinity >5 ppt. Improve aeration and bottom management. Use feed with adequate mineral content.

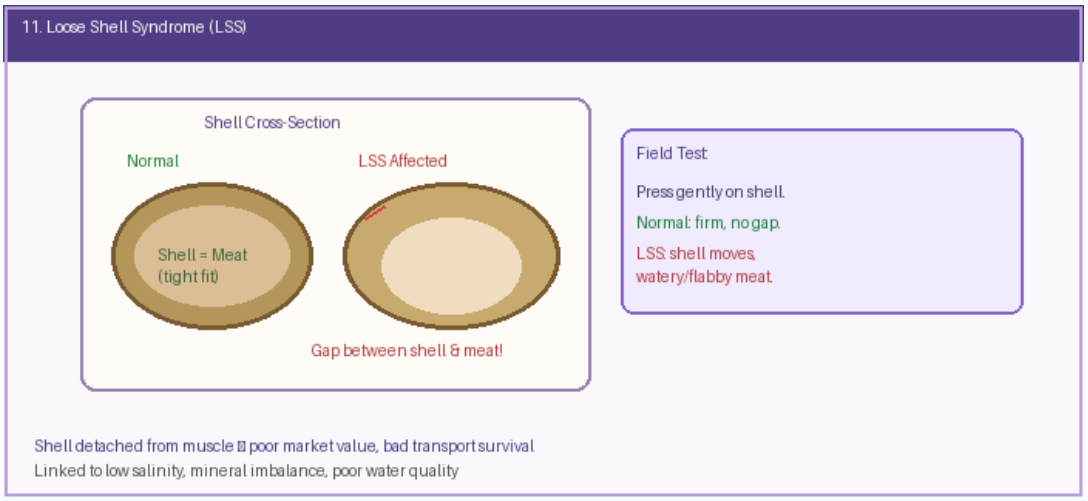

Fig 11: Shell cross-section — normal tight shell-to-meat fit (left) vs LSS-affected shrimp showing gap between shell and atrophied meat (right)

12. Hepatopancreatic Parvovirus (HPV) [Moderate Risk]

Cause / Type	Viral (Parvovirus). Primarily affecting larval and juvenile stages. Persistent in Nellore district hatcheries and farms.
Symptoms	Significantly stunted growth. Pale, shrunken, yellowish hepatopancreas. Shell may appear cloudy or rough. Does not typically cause acute mass mortality.
Behavior	No distinctive behavioral abnormality; affected shrimp appear externally normal but eat less and grow slowly.
Economic Impact	20-40% of the pond population significantly undersized at harvest despite normal feeding inputs. Hidden economic cost in poor FCR.
Prevention / Management	PCR screening of broodstock and PL at hatchery stage. No direct treatment available. Maintain excellent water and sediment quality to reduce secondary disease pressure.

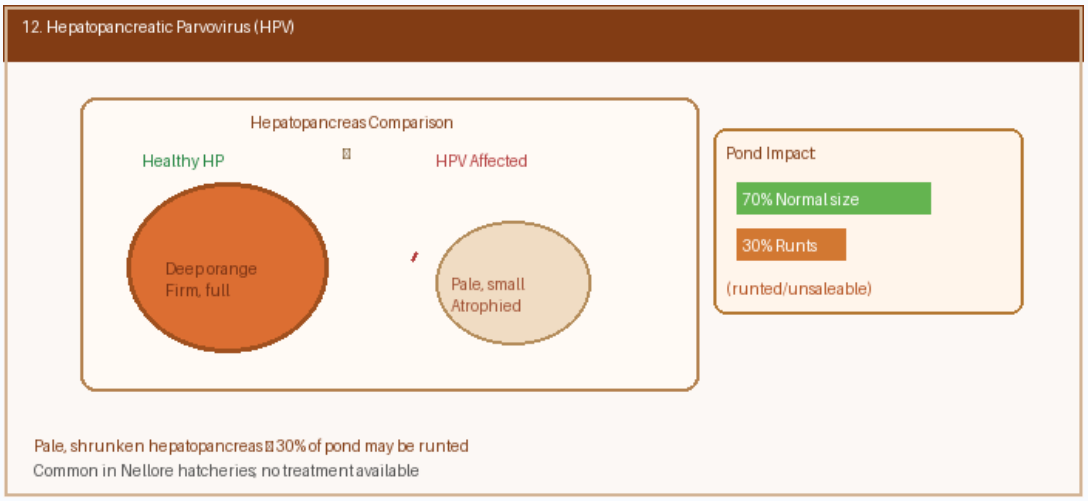

Fig 12: Hepatopancreas comparison — healthy deep-orange firm tissue (left) vs HPV-affected pale, shrunken atrophied hepatopancreas (right). Pond impact: ~30% of shrimp become runts

13. Yellow Head Disease (YHD) [High Risk]

Cause / Type	Viral (Okavirus genus, Roniviridae family). Outbreaks seasonal — often during summer-to-monsoon transition when parameters fluctuate sharply.
Symptoms	Distinct yellow/golden discoloration of cephalothorax (head-thorax area). Gills appear pale yellow, brown, or white. Pale yellow hepatopancreas.
Behavior	Dangerous biphasic pattern: voracious feeding for 2-3 days, then abrupt cessation followed by mass die-off at pond surface. This pattern is a key field indicator.
Economic Impact	Mortality can reach 100% within 3-5 days of onset. Speed of progression makes control difficult.
Prevention / Management	Monitor feeding response closely during seasonal transitions. Maintain stable water conditions. PCR screen broodstock. Reduce pond stress during monsoon changeover.

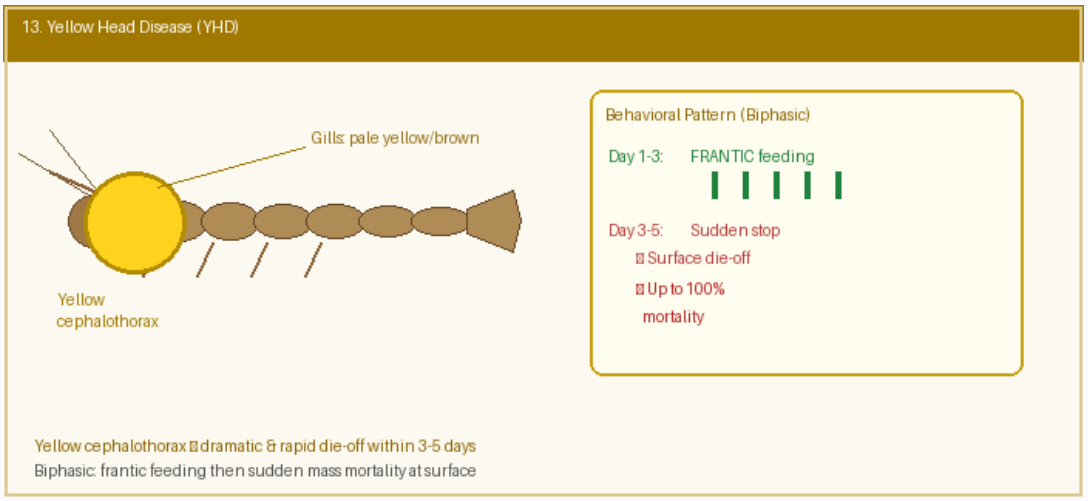

Fig 13: YHD — yellow cephalothorax (head area) is the primary visual sign. Timeline shows the diagnostic biphasic pattern: frantic feeding (Days 1-3) then sudden die-off (Days 3-5)

14. Luminescent Vibriosis — Vibrio harveyi [High Risk]

Cause / Type	Bacterial. Specifically caused by Vibrio harveyi, a bioluminescent strain. Most damaging in hatcheries and early pond stages (Days 1-30). Thrives in warm water (>28 degrees C) with high organic load.
Symptoms	Greenish-blue luminescence (glow) visible in pond water and on infected shrimp at night. Black spots on body. Cloudy or opaque muscle. High mortality in larval and PL stages.
Behavior	Infected larvae may display spinning, erratic swimming behavior before dying.
Economic Impact	Can cause near-total hatchery losses if uncontrolled. In ponds, outbreaks during first 30 days post-stocking are especially destructive.
Prevention / Management	Reduce organic load in hatchery and pond water. Apply probiotics (Bacillus, Lactobacillus) to outcompete Vibrio. Disinfect water intake. Avoid antibiotic overuse which creates resistance.

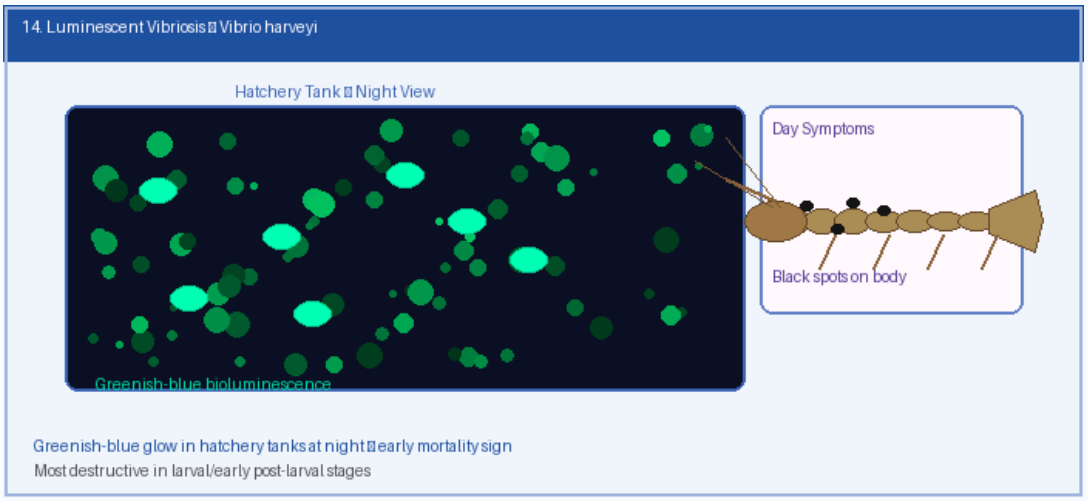

Fig 14: Hatchery tank at night showing greenish-blue bioluminescence from V. harveyi infection (left). Day symptoms: black spots on shrimp body (right inset)

15. Tail Necrosis (Rotten Tail) [Moderate Risk]

Cause / Type	Bacterial. Opportunistic Vibrio spp. colonizing physically damaged tail tissue. Prevalent in intensive ponds with poor bottom management and high sludge accumulation.
Symptoms	Uropods (tail fan edges) turn black and erode, progressing inward. In advanced cases, the entire tail fan may be partially destroyed. Secondary bacterial septicemia may follow.
Behavior	Affected shrimp swim erratically due to loss of tail function; vulnerable to cannibalism.
Economic Impact	Reduces market value significantly — tail-damaged shrimp rejected or heavily discounted at factories. Indicates systemic pond health failure.
Prevention / Management	Primary: pond floor management — increase aeration, apply probiotics, reduce stocking density if >150,000/acre. Improve water quality. Minimize unnecessary disturbance to reduce physical injury.

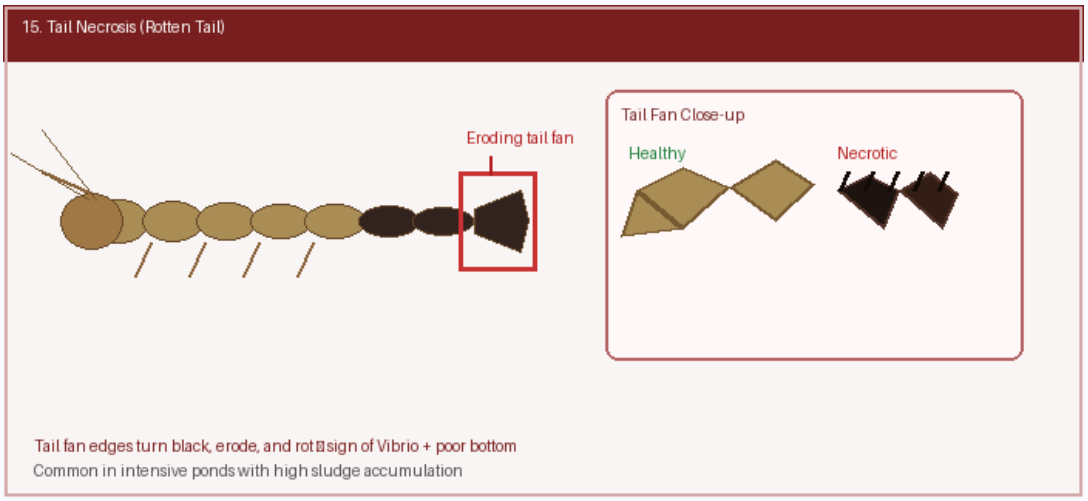

Fig 15: Tail necrosis progression — healthy tail fan (right, light brown) vs necrotic black eroding uropods (left). Caused by Vibrio on damaged tissue in high-sludge ponds

General Disease Prevention & Best Practices

General Best Practices for Disease Prevention

- ✓ Biosecurity First: Use certified SPF (Specific Pathogen Free) seed from accredited hatcheries. Screen broodstock for all major pathogens before stocking.
- ✓ Water Quality Monitoring: Test pH (7.5-8.5), DO (>4 mg/L), ammonia (<0.1 mg/L), salinity, and temperature daily. Sudden fluctuations trigger most viral outbreaks.
- ✓ Pond Bottom Management: Use paddle wheels/aerators to prevent sludge accumulation. Apply probiotics (Bacillus spp.) to reduce organic load and toxic gas (H2S, NH3) buildup.
- ✓ Feeding Discipline: Remove uneaten feed promptly. Overfeeding is the #1 driver of poor water quality and Vibrio proliferation.
- ✓ Early Warning System: Use check trays daily. A 0.5% daily mortality rate is a warning sign — investigate and respond within 24 hours.
- ✓ Quarantine Protocol: Never introduce water, equipment, or personnel from infected ponds without disinfection (chlorine 30 ppm).
- ✓ Record Keeping: Maintain daily logs of mortality counts, feed consumption, and water parameters to detect disease onset early.

Field Diagnostic Quick Guide

What You Observe in the Field ->	Likely Disease(s) to Investigate
White spots on shell	WSSV — confirm by PCR
Stunted growth, size variation	EHP or IHHNV/RDS — PCR and histopath
White floating feces	White Feces Syndrome (WFS) — check EHP also
Glow at night (water or shrimp)	Luminescent Vibriosis (V. harveyi)
Opaque white tail muscle	IMNV (Myo) or CMNV — PCR both
Yellow/golden head area	Yellow Head Disease (YHD)
Blue/purple iridescent glow	DIV1 — Mandatory reportable, call MPEDA
Slow, continuous daily deaths (~1%)	Running Mortality Syndrome (RMS)
Black/eroding gills	Black Gill Disease — check pond bottom quality
Shell loose, watery muscle	Loose Shell Syndrome (LSS)
Bent rostrum, deformity	IHHNV / Runt Deformity Syndrome
Reddish body, red appendages + glow	Vibriosis (Bacterial Septicemia)
Black, eroding tail fan edges	Tail Necrosis — Vibrio-driven
Voracious feeding then sudden death	Yellow Head Disease (YHD)
Shrimp die at bottom, not visible	CMNV (Covert Mortality Nodavirus)

Important Disclaimer

This document is compiled for field reference and educational purposes. Disease illustrations are scientific diagrams for educational use only and do not replace photographic diagnosis. Disease prevalence data reflects conditions as of 2025-2026. All outbreak responses and treatment decisions should be made in consultation with a licensed aquaculture veterinarian or pathologist. Reportable diseases (WSSV, DIV1, YHD, IMNV) must be notified to relevant authorities (MPEDA, DoF, NFDB).